MILITARY SPECIFICATION

INDICATOR, VERTICAL SPEED, 3-INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of vertical speed indicator, range 0 to 6,000 feet per minute (fpm) climb and descent.

*2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-N-3336	Nuts, Self-Locking, Clip-In Type, Instrument Mounting
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-L-25142	Luminescent Material, Fluorescent
MIL-P-26514	Polyurethane Foam Rigid or Flexible for Packaging

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STANDARDS

Federal

FEDERAL STANDARD NO. 595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MS28049	Indicator, Vertical Speed, 3 Inch
MS28105	Window, Dial - Aircraft Instrument Cover, Glass
MS33558	Numberals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Data. Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2 herein (see 6.2).

3.3 Materials. Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications shall be of the best commercial quality and suitable for the purpose intended.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the instrument, except where magnetic materials are essential.

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3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.4.1 and 3.4.2.

3.4.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.4.2 Standard parts. With the exception indicated in 3.4, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.5 Design and construction. The indicator shall conform to MS28049 with the design being reasonably simple to permit overhaul or repair without requiring numberous special tools and fixtures. Special tools shall be subject to the approval of the procuring activity. Indicators shall be constructed to withstand the normal strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.5.1 Case. The case shall consist of a body and bezel ring and shall conform to MS28049. The case shall be airtight, except for the static pressure connection.

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3.5.1.1 Body. The body shall be made of nonferrous, low-density metal, shall be uniform in texture, and shall have a smooth surface with durable lusterless black finish, color No. 37038 of Federal Standard No. 595. The mounting flange portion of the body shall have a dull finish.

3.5.1.2 Bezel ring. The bezel ring shall be made of nonferrous, low-density metal and shall have a durable lusterless black finish, color No. 37038 of Federal Standard No. 595. The bezel ring shall be held in place by means of screws. Provision shall be made for the replacement of the cover glass by removal of the bezel ring.

3.5.1.3 Cover glass. The cover glass shall conform to MS28105.

3.5.1.4 Operating mechanism. The operating mechanism shall be of the leak type. The operating mechanism shall be sufficiently rugged to prevent small distortion of the case from affecting the accuracy of the instrument.

3.5.1.5 Zero-setting system. A zero-setting system, controlled by a slotted setting shaft or screw, shall be provided that will permit the instrument to be set to indicate zero.

3.5.1.5.1 The use of the zero-setting system shall not change the calibration of the instrument.

3.5.1.6 Dial-to-cover-glass distance. The distance from the dial to the cover glass shall be as small as practicable and shall not exceed 1/4 inch.

3.5.1.7 Pointers. The pointers shall conform to MS33585-3, except that the distance from the center of the hub to the tip of the tapered end shall be 1.218 $\pm$ 0.010 inch. They shall be light and sufficiently rigid to prevent oscillation under vibration and shall be firmly attached to the mechanism. Suitable stops shall be incorporated within the mechanism to limit pointer movement. The pointer shall rotate to the 6,000-foot graduation +1/16 inch on both ascent and descent scales.

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3.6 Performance. The indicator shall conform to the following requirements when subjected to the applicable test specified in section 4:

(a) Temperature: Temperatures over the range of -65° C to +71° C.
(b) Vibration error: Vibration error at an amplitude between 0.003 and 0.005 inch at frequencies of 500 to 3,000 cycles per minute (cpm).
(c) Vibration and endurance: Vibration endurance at a circle diameter amplitude of between 0.018 and 0.020 inch and a frequency to be varied uniformly from 500 to 3,000 cpm.
(d) Humidity: Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of both water and frost.
(e) Fungus: Fungus growth as encountered in tropical climates.
(f) Salt spray: Exposure to salt sea atmosphere.
(g) Altitude: Altitude in the range of sea level to 50,000 feet (3.436 inches mercury (Hg)).
(h) Leakage: Leakage not to exceed 0.05 inch Hg when subjected to a vacuum of 25 inches Hg.
(i) Scale error: Scale error (see 6.4) not to exceed the limits specified in 4.7.3.
(j) Pointer lag: Pointer lag shall be not less than 7 seconds or more than 12 seconds when subjected to the test specified in 4.7.4.
(k) Overpressure: Overpressure within the range specified in 4.7.11.
(l) Position error: Position error which shall not exceed 50 fpm.
(m) Magnetic effect: Magnetic effect not to exceed 1 degree in a magnetic field intensity of 0.17 to 0.19 oersted.
(n) Mounting lugs: Each mounting lug shall withstand a load of 175 pounds for a period of 1 minute.

*3.7 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-STD-100 shall govern changes in the manufacturer's part numbers.

3.8 Finish and protective coating.

3.8.1 Aluminum-alloy parts. All aluminum-alloy parts, except small holes, pipe threads, case inserts, and the dial, if of aluminum or aluminum alloy, shall be covered with an anodic film conforming to MIL-A-8625.

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3.8.2 Steel parts. Steel parts shall be cadmium plated where practicable in accordance with type I, class 2 of QQ-P-416.

3.9 Screw threads. The threads of all machine screws shall conform to MIL-S-7742.

3.10 Weight. The indicator shall be weighed completely assembled, without shipping plug and mounting screws, and shall not exceed 1.75 pounds.

3.11 Markings.

3.11.1 The dial and pointers shall be marked as shown on figure 1.

3.11.2 The dial letters and numerals shall conform to MS33558.

3.11.3 Fluorescent-luminescent material shall conform to type I or III of MIL-L-25142, as applicable.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Installation.

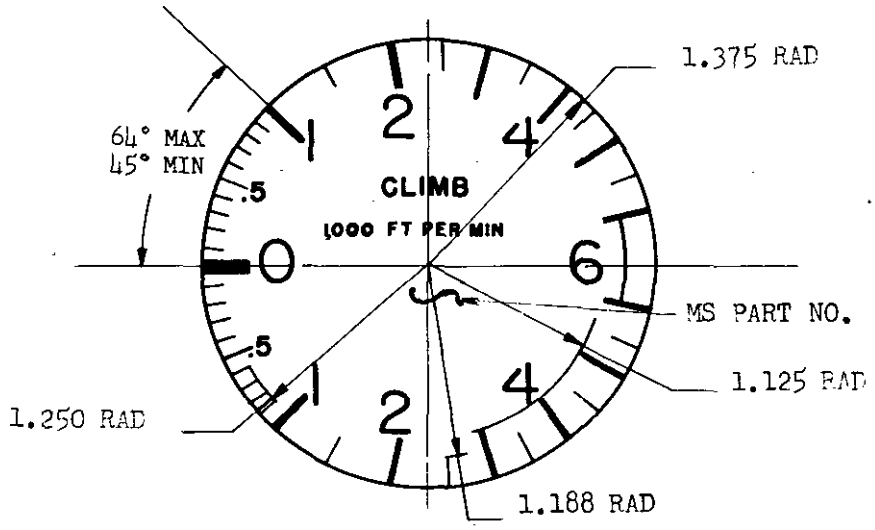
3.13.1 Mounting screws. The contractor shall furnish sufficient mounting screws, having durable dull black oxidized or black nickel finish. Length shall be sufficient for mounting on a panel 3/16 inch thick.

3.13.2 Removable spring nuts. Unless otherwise specified, a sufficient number of removable No. 6-32UNC-2 spring nuts to fit the mounting lugs of the instrument, of a type conforming to MIL-N-3336 shall be furnished.

3.13.3 Envelope. When required, the contractor shall furnish and pack with each instrument, an envelope containing mounting screws and spring nuts which shall be marked with the following information:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS AND SPRING NUTS

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NUMERALS SHALL DISTINCTLY INDICATE THE GRADUATION TO WHICH EACH APPLIES

MARKING	HEIGHT OR LENGTH	WIDTH OF LINE	MATERIAL OR FINISH
ZERO GRADUATION	------	0.062	LUSTERLESS WHITE, COLOR NO. 37875 OF FEDERAL STANDARD NO. 595 OR FLUORESCENT LUMINESCENT MATERIAL (SEE 6.2(e)).
1,000 FT-GRADUATIONS	------	.031	
100 FT-GRADUATIONS	------	.020	
500 FT-GRADUATIONS	------	.020	
ARC CONNECTING 6,000 FT	------	.031	
MARKS	------	.031	
NUMERALS .5	0.125	----	
NUMERALS 0, 1, 2, 4, 6	.312	----	
Lettering "CLIMB"	.125	----	
SHADED PART OF POINTER	-----	----	
LETTERING "1,000 FT PER MINUTE".	.062	.016	LUSTERLESS WHITE, COLOR NO. 37875 OF FEDERAL STANDARD NO. 595.
(MS PART NO.)	.063	----	LUSTERLESS BLACK, COLOR NO. 37038 OF FEDERAL STANDARD NO. 595.
UNSHADED PART OF POINTER AND BACKGROUND DIAL.	-----	----	

FIGURE 1. *Dial marking.*

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3.14 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

3.14.1 Dimensions. Dimensions and tolerances not specified herein shall be as close as is consistent with the best shop practices. Where dimensions and tolerances affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (4.3)
(b) Quality conformance inspection (4.4)

*4.3 Qualification inspection.

4.3.1 Sampling instructions. Qualification test samples shall consist of three indicators manufactured in accordance with this specification. The indicators submitted for qualification testing shall not have been previously tested, except for the individual tests (4.4.1). The samples shall be forwarded to the testing laboratory designated in the letter of authorization (see 6.3).

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4.3.1.1 Sample identification. The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for qualification tests
INDICATOR, VERTICAL SPEED, 3-INCH
Submitted by (manufacturer's name, date) for qualification tests in accordance with MIL-I-7721C under authorization (reference letter authorizing tests)
MS28049
Manufacturer's part No.

4.3.2 Qualification required. Prior to actual procurement, the indicator covered by this specification shall pass the qualification tests specified herein.

*4.3.3 Qualification inspection. The qualification inspection shall consist of all the tests specified under 4.6 and 4.7.

*4.4 Quality conformance inspection. The quality conformance inspection shall consist of individual tests, examinations, and sampling plans A and B tests (4.4.1, 4.6, and 4.4.2, respectively).

4.4.1 Individual tests. Each indicator shall be subjected to the following examinations and tests, as described under 4.6 and 4.7:

(a)	Examination of product	(4.6.1)
(b)	Zero setting	(4.7.1)
(c)	Leakage	(4.7.2)
(d)	Scale error at room temperature	(4.7.3)
(e)	Position error	(4.7.13)

4.4.2 Sampling plans A and B tests.

4.4.2.1 Sampling plan A tests. Five indicators selected at random from each 500 or less on the contract or order shall be subjected to the following tests, as described under 4.6 and 4.7:

(a)	Individual tests	(4.4.1)
(b)	Pointer lag	(4.7.4)
(c)	High temperature	(4.7.5)
(d)	Low-temperature zero error	(4.7.6)
(e)	Scale error during temperature change	(4.7.7)
(f)	Low-temperature scale error	(4.7.8)
(g)	Zero error during temperature change	(4.7.9)
(h)	Vibration error	(4.7.10)
(i)	Overpressure	(4.7.11)

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4.4.2.1.1 Rejection. When one or more items from a sampling plan A lot fails the specified tests, this shall be cause for rejection of the lot.

4.4.2.2 Sampling plan B tests. Unless otherwise specified, 3 indicators selected at random from the first 10 items of the contract or order shall be subjected to the following tests, as described under 4.6 and 4.7:

(a)	Sampling plan A tests	(4.4.2.1)
(b)	Vibration endurance	(4.7.12)
(c)	Position error	(4.7.13)
(d)	Magnetic effect	(4.7.14)
(e)	High altitude-low temperature	(4.7.15)
(f)	High-temperature exposure	(4.7.16)
(g)	Humidity	(4.7.17)
(h)	Fungus	(4.7.18)
(i)	Salt spray	(4.7.19)
(j)	Mounting lugs	(4.7.20)

4.4.2.2.1 Rejection. When one item from a sampling plan B lot fails the specified tests, this shall be cause for rejection of the lot.

4.5 Test conditions.

4.5.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.5.2 Vibration stand. A vibration stand shall be used which will vibrate at any desired frequency at 500 to 3,000 cpm and subject the indicator to vibration such that a point on the indicator case will describe, in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude (see 6.4) as specified herein.

4.5.3 Tapping. Unless otherwise specified, the instrument shall be tested in normal operating position and shall be lightly tapped (see 6.4) before a test reading is taken.

4.6 Examinations.

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4.6.1 Examination of product. Each indicator shall be inspected to determine compliance with this specification with respect to material, workmanship, and general requirements.

4.6.2 Packaging, packing, and marking. Preparation for delivery shall be examined for conformance to section 5.

4.7 Test methods.

4.7.1 Zero setting. The adjusting screw shall be moved throughout its range in both the up and down directions. The adjustable range of pointer movement shall be not less than 400 fpm up to not less than 400 fpm down.

4.7.2 Leakage. The static pressure connection of the indicator shall be suitably connected to a mercury manometer and sources of vacuum and pressure. A vacuum of 25 inches Hg shall be applied to the static pressure connection at a rate not exceeding 20,000 fpm. With the source closed off, the difference between the mercury levels in the manometer during a period of 1 minute shall not exceed 0.05 inch Hg. The same procedure shall then be repeated with a pressure of 10 inches Hg. The same tolerance shall apply.

4.7.3 Scale error at room temperature. The indicator, with the pointer carefully set to zero, shall be placed in an altitude chamber. A mercury barometer shall be connected to the chamber. The pressure shall be reduced at a rate such that the pointer indicated 500 fpm. While the pointer indicates 500 fpm, the time required for the mercury barometer to change from a pressure corresponding to 1,000 feet altitude to that of an altitude of 1,500 feet shall be determined by a stop watch. The pressure shall then be increased at such a rate that the pointer will indicate a descent of 500 fpm and the time required for the barometer to change from 1,500 feet altitude to 1,000 feet altitude determined. This procedure shall be continued for the test rates and altitudes shown in table I. The scale error shall not exceed the tolerances listed in table I.

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TABLE I. Scale error at room temperature

Std. altitude interval (ft.)	Pressure (in. of Hg)	Test rate (fpm)	Tolerance scale error (fpm)	
			Up	Down
1,000 to 1,500	---	500	100	100
1,000 to 2,000	---	1,000	200	200
2,000 to	27.82	2,000	300	300
4,000	25.83	3,000	300	300
		4,000	400	400
		5,000	500	500
15,000 to	---	2,000	300	300
17,000	---	4,000	400	400
28,000 to	---	2,000	300	300
30,000	---	4,000	400	400

4.7.3.1 When testing several instruments in one chamber at the same time, one instrument may be selected to indicate the test rate. This test rate shall be maintained at a constant value while the test readings (see 6.4) of all the instruments are observed and noted, and while the time and altitude interval on the barometer are noted. The computed rate (see 6.4) shall be obtained. The scale errors shall be determined.

4.7.4 Pointer lag. The static pressure connection of the indicator shall be suitably connected to a water manometer and alternate sources of vacuum or pressure. A vacuum shall be carefully applied to cause the pointer to indicate a rate of ascent of approximately 2,500 fpm. The vacuum shall then be instantaneously released by opening the static pressure to the atmosphere, causing the pointer to indicate a rate of descent greater than 2,000 fpm. As the pointer moves back toward zero, a stop watch shall be started at the instant the pointer passes the 2,000-fpm dive graduation and shall be stopped when the pointer passes the 200-fpm dive graduation. The elapsed time shall be recorded. This procedure shall be repeated, except that a pressure shall be carefully applied to the static pressure connection and the time determined for the indication to change from 2,000 fpm climb to 200 fpm climb. The lag of the pointer shall be not less than that specified in 3.6(j) for each procedure. During this test and at any time after the vacuum (or pressure) is released, the pointer shall move smoothly without excessive friction.

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4.7.5 High temperature. The indicator shall be subjected to a temperature of 71° C for a period of 3 hours. It shall then be allowed to cool to room temperature after which it shall be subjected to the tests for scale error at room temperature (4.7.3) and case leakage (4.7.2). The indicator shall conform to the requirements specified for scale error at room temperature for the interval and rates listed in table II. During and after this test, the setting system shall operate satisfactorily.

TABLE II. Scale error during temperature tests

Std. altitude interval (ft.)	Test rate (fpm)	Tolerance scale error (fpm)	
		Up	Down
2,000 to 4,000	2,000	300	300
	2,000	400	400
15,000 to 17,000	2,000	300	300
	4,000	400	400
28,000 to 30,000	2,000	300	300
	4,000	400	400

4.7.6 Low-temperature zero error. Before starting this test, the pointer shall be carefully set to zero at room temperature. The indicator shall then be placed in a test chamber and cooled to approximately -54° C within any length of time between 15 minutes and 5 hours and allowed to remain at the temperature for a period of not less than 3 hours. With the test chamber at atmospheric pressure, a low-temperature zero error during the cooling period and during the 3-hour period at 54° C shall not exceed 150 fpm.

4.7.7 Scale error during temperature change. At intervals of 1 hour during the cooling period and during the 3-hour period at -54° C, the indicated rates (see 6.4) of the indicator corresponding to the test rates of 2,000 fpm climb and descent, in the standard altitude interval between 2,000 and 4,000 feet, shall be obtained. These indicated rates, corrected for the corresponding zero errors, shall not exceed the values listed in table II.

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4.7.8 Low-temperature scale error. After a temperature of -54° C has been maintained within the chamber for 3 hours, the indicator shall be subjected to the test specified for scale error at room temperature, except that the intervals and rates listed in table II shall be used. The indicated rates, corrected for the low-temperature zero error, at the end of the 3-hour period at -54° C shall not exceed the values listed in table II. During and after this test, the setting system shall operate satisfactorily.

4.7.9 Zero error during temperature change. After the low-temperature scale error test has been performed, the indicator shall be allowed to return to room temperature at atmospheric pressure by immediate removal from a temperature of -54° C to room temperature. While the temperature of the indicator is increasing, the indication shall be observed for a period of 1/2 hour, at 1-minute intervals for the first 5 minutes and at 5-minute intervals thereafter. During the first 5-minute interval, the maximum reading shall not exceed 300 fpm; thereafter, the maximum reading shall not exceed 200 fpm.

4.7.10 Vibration error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane, and with the plane of the mounting lugs of the indicator vertical. For this test, the pointer shall be set to zero. No pressure or vacuum shall be applied during the test. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies of 500 to 3,000 cpm in order to determine that the natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated, the maximum range of pointer oscillation at any time shall not exceed 50 fpm. The pointer variation from the zero position shall not exceed 50 fpm.

4.7.11 Overpressure. With the indicator in an altitude test chamber after the pointer has been carefully set to zero, the pressure shall be decreased at a rate of 60,000 fpm until a pressure corresponding to 50,000 feet is reached. The pressure shall then be increased at a rate of 60,000 fpm until atmospheric pressure is reached. Five minutes later the indication of the pointer shall be within 100 fpm of it original indication.

4.7.12 Vibration endurance. The indicator shall be mounted on a vibration stand as specified for the vibration error test with the pointer set at zero. The indicator shall then be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 500 to 3,000 cpm and return once each hour during this test. The difference between the indication of the pointer

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at the beginning and at the end of the 3-hour vibration period shall not exceed 50 fpm. In addition, no looseness in the mechanism nor damage to any part of theindicator shall result from the vibration.

4.7.13 Position error. The pointer shall be set to zero. With atmospheric pressure applied, readings shall be taken while the instrument is being tapped lightly in each of several different positions. The change in indication shall not exceed 50 fpm.

4.7.14 Magnetic effect. The indicator shall be held in various positions with its nearest part 5 inches from and magnetically east or west from the center of a short bar magnet compass. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass shall not exceed 1 degree.

4.7.15 High altitude - low temperature. The indicator shall be placed in an altitude chamber maintained at a temperature of -65° C and the pressure reduced to simulate an altitude condition of 50,000 feet (3.436 inches Hg absolute pressure) for a period of 4 hours. The indicator shall then be maintained at a temperature of -65° C, atmospheric pressure, for an additional period of 44 hours. After returning to room temperature, the indicator shall be carefully examined for any evidence of damage. No damage to any part of the indicator shall result from this test.

4.7.16 High-temperature exposure. The indicator shall be subjected to a temperature of 71° ± 2° C for a period of 24 hours. At the end of the 24-hour period and while still at the high temperature, the indicator shall operate without indication of sticking or erratic action. After the chamber has been returned to room temperature and the indicator has saturated at room temperature for a minimum of 4 hours, it shall be subjected to and pass the individual tests (4.4.1). There shall be no damage which would adversely affect subsequent operation.

*4.7.17 Humidity. The indicator shall be subjected to humidity test described in paragraph 4.7.17.1. To simulate installed conditions, the external connections to the indicator shall be connected with 10 feet of 1/4-inch copper or aluminum tubing coiled so that moisture can drain out of the open end. Within 1 hour after the 360-hour period, the indicator shall be subjected to and pass the scale error at room temperature test (4.7.3).

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*4.7.17.1 Humidity test procedure. The equipment shall be placed in the test chamber and set up to simulate installed conditions. The test chamber shall be vented to the atmosphere to prevent the build-up of pressure. Prior to the starting of the test period, the chamber temperature shall be between 20° and 38° C (68° and 100° F) with uncontrolled humidity. During the first 2-hour period, the temperature shall be gradually raised to 71° C (160° F) unless otherwise specified in detail test requirements. The temperature selected shall be maintained during the next 6-hour period. The velocity of the air throughout the test area shall not exceed 150 feet per minute. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20° to 38° C (68° to 100° F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. Steam or distilled water having a pH value between 6.5 and 7.5 at 25° C (77° F) shall be used to obtain the desired humidity. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles) or as specifed by the applicable equipment specification. At the conclusion of the 240-hour period, the equipment shall be returned to standard conditions. The equipment shall then be operated and inspected within 1 hour for normal operating conditions and tolerances listed in paragraph 3.6.

*4.7.18 Fungus. The indicator shall be subjected to fungus resistance test described in paragraph 4.7.18.1. The installed conditions of the humidity test (4.7.17) shall apply.

*4.7.18.1 Fungus test procedure. Four groups of fungi are listed below and one species of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at a temperature between 22° C and 32° C (72° F and 89° F) shall be utilized. Approximately 10 ml. of the solution shall then be introduced directly into a tube culture of the fungus and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an innoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions from the four types of fungi shall be mixed together to provide a composite suspension. Actively growing cultures between 7 to 21 days old after initial innoculation shall be used for the preparation of the spore suspension. After preparation, the spore suspension will not be kept for more than a 24-hour period at temperatures from 22° C to 32° C (72° F to 89° F), or not more than 48 hours at temperatures from 2° C to 7° C (35° F to 45° F). The equipment, including applicable external connections, shall be placed in a mold chamber described in Appendix A of this document, (reference paragraph 6.7.1) maintaining an

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internal temperature of 30° ± 2° C (86° ± 3.6° F) and a relative humidity of 95 ± 5 percent, and aprayed with the suspension of mixed spores. The test period shall be 14 days. At the end of the test, the parts shall be visually examined for normal conditions. The equipment shall be operated and inspected for normal conditions and tolerances listed in paragraph 3.6.

Organism		
Group I	Chaetomium globosum	6205
	Myrothecium verrucaria	9095
Group II	Memenoniella echinata	11973
	Aspergillus niger	6275
Group III	Aspergillus tamarii	10836
	Aspergillus terreus	10690
Group IV	Penicillium citrinum	9849
	Penicillium ochrochloron	9112

NOTE: When ordering cultures, request the cultures by name and serial number. The serial numbers of the cultures remain the same regardless of source. One known source is listed below. Other sources are acceptable.

Source: American Type Culture Collection
12301 Parklawn Dr., Rockville, MD 20852

*4.7.19 Salt spray. The indicator shall be subjected to salt spray test provided in paragraph 4.7.19.1 for a period of 50 hours. The installed conditions of the humidity test (4.7.17) shall apply. At the end of the 50-hour period, the indicator shall be subjected to the individual tests (4.4.1).

*4.7.19.1 Salt spray test procedure. The procedure shall be in accordance with Appendix B of this specification. The length of the salt spray test shall be not less than 50 hours. At the end of the test period, the equipment shall be operated and inspected within 1 hour. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water prior to operation. Equipment shall again be operated and inspected for conditions and tolerances provided in paragraph 4.4.1 48 hours later.

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4.7.20 Mounting lugs. The indicator case, with mechanism removed, shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting nut and attached to a pull strap. A load of 175 pounds shall be applied along the axis toward the front of the case for a period of 1 minute. There shall be no breakdown.

*4.8 Inspection and test. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to assure compliance with Section 5 of this specification.

*5. Preparation for delivery. Unless otherwise specified in the individual contract or order, the following shall apply:

*5.1 Preservation and packaging.

*5.1.1 Level A. (Reference definition in MIL-STD-794) Indicators shall be packaged in accordance with Method IIC of MIL-P-116, one each in a unit container, conforming to PPP-B-636, Weather Resistant Class. Each unit shall be cushioned with MIL-P-26514 Polyurethane. Cushioning thickness, density and configuration shall insure shock is dampened below the maximum permissible nonoperating shock limitations established for the indicator. Completed packs shall be afforded the rough handling tests prescribed by MIL-STD-794 to confirm packaging reliability.

*5.1.2 Level C. (Reference definition in MIL-STD-794) Indicators shall be packaged identically to Level A requirements, except Method III (MIL-P-116) applies in lieu of Method IIC and domestic class, PPP-B-636 Box, shall be used as the unit container.

*5.2 Packing.

*5.2.1 Level A. Indicators packaged, as specified in 5.1.1, shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class. Insofar as practicable, shipping containers shall be uniform shape, size, of minimum tare and cube consistent with the protection required.

*5.2.2 Level B. Indicators packaged, as specified in 5.1.1, shall be packed in PPP-B-636 domestic class boxes. Other requirements specified in 5.2.1 shall apply.

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*5.2.3 Level C. Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirement of this level.

*5.3 Marking.

*5.3.1 Container markings. Unit package and shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

"Indicator Vertical Airspeed, 3-Inch"

6. NOTES

6.1 Intended use. The vertical speed indicators are intended for use on aircraft to indicate the vertical component of the aircraft's speed based on the rates of change of atmospheric pressure.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether data required (see 3.2).
(c) Levels of packaging and packing (see 5.1 and 5.2).
(d) Installation instruction sheets (see 6.2.1).
(e) For Army procurement only, whether fluorescent luminescent material is required.

6.2.1 Installation instructions. If required, the contractor should furnish with each indicator one printed copy of instructions, with illustrations and diagrams, if necessary, covering the installation of the indicator. Prior to printing, two copies should be furnished to the procuring activity for approval. Instructions should be printed on 8-1/2- by 11-inch paper or 17- by 11-inch paper.

*6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification.

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The activity responsible for the Qualified Products Lists is Oklahoma City ALC/MMEO, Tinker AFB, OK 73145 and information pertaining to qualification of products may be obtained from that activity.

6.4 Definitions.

6.4.1 Amplitude. Whenever the word "amplitude" is specified, it means the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Tapping. The phrase "tapped lightly" is the application of a light vibration sufficient to overcome that amount of residual friction normally associated with good quality aircraft instruments.

6.4.3 Test reading. "Test reading" is the indication of the instrument during the test.

6.4.4 Test rate. "Test rate" is the rate specified in column 3 of tabe I and column 2 of table II at which the instrument is to be tested.

6.4.5 Computed rate. "Computed rate" is the rate computed by dividing the altitude interval (2,000 feet as measured by a barometer) by the time in seconds (required for the change in altitude) and multiplying by 60.

$$\text{Computed rate} = \frac{\text{Altitude interval X 60}}{\text{Time in seconds}}$$

6.4.6 Computed rate correction. "Computed rate correction" is the algebraic difference between the test rate and the computed rate.

Example 1.	Test rate	+ 2,000
	Computed rate	+ 1,925
	Computed rate correction	+ 75
Example 2.	Test rate	+ 2,000
	Computed rate	+ 2,025
	Computed rate correction	- 25

6.4.7 Indicated rate. "Indicated rate" is the reading of the instrument as computed herein to correspond to a "test rate." The indicated rate of the instrument is computed by adding algebraically the test reading of the instrument and the computed rate correction.

6.4.8 Scale error. "Scale error is the difference between the "test rate" and the "indicated rate."

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*6.5 "The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

6.6 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ASCC AIR STD 10/8 and STANAG 3320. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

*6.7 Cross reference information. References to MIL-E-5272 contained in previous editions of MIL-I-7721B have been deleted in this edition because MIL-E-5272 has been made inactive for new design. Test procedures contained in MIL-E-5272 that are needed in MIL-I-7721B have been transposed to this document with a minimum of changes. It is the intent of these changes that indicators qualified under MIL-I-7721B are acceptable under MIL-I-7721C; new tests are not required.

*6.7.1 Interchangeability information for mold chamber. Appendix A to this document contains information formerly covered by Section 3 of MIL-C-9452 which has been rescinded. Small changes have been made to update essential data. (Reference paragraph 4.7.18.1)

*6.7.2 Interchangeability information for salt spray test. Appendix B to this document contains information formerly contained in FED-STD-151, Procedure 811, which has been rescinded. Small changes have been made to update essential data. (Reference paragraph 4.7.19.1)

*6.7.3 Availability of documents. Copies of rescinded specifications such as MIL-E-5272, MIL-C-9452 or FED-STD-151, Procedure 811, may be obtained from:

Commanding Officer
Naval Publications and Forms Center
5801 Tabor Avenue
Philadelphia, PA 19120

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Navy - AS
Air Force - 71

Preparing activity:
Air Force - 71

Project No.
6610-0372

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APPENDIX A

10. FUNGUS RESISTANCE TESTING CHAMBER

10.1 Scope. This appendix outlines the design, fabrication, and testing to provide a facility for the fungus resistance testing of aeronautical and associated equipment formerly covered by MIL-C-9452 which has been cancelled.

10.2 Requirements.

10.2.1 General. The facility shall consist of an insulated chamber lined with a corrosion, moisture, and fungus-proof material such as stainless steel. The shelves shall be removable and fabricated from perforated stainless steel or wire capable of supporting 75 pounds per square foot. A humidifier, heat source, and other equipment necessary to meet the requirements outlined in this appendix and in Specification MIL-I-7721 regarding fungus resistance testing shall be provided. The facility shall be constructed so that it can be readily cleared and sterilized.

10.2.2 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

10.2.2.1 Standard parts. AN and MIL standard parts shall be used whenever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

10.2.3 Design. This fungus resistance testing chamber shall be a self-contained unit capable of establishing, maintaining, and withstanding the environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-7721. The facility shall be designed to meet the performance requirements outlined in 10.2.5.

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10.2.3.1 The chamber interior shall be completely fungus-proof and the chamber bottom shall be leak-proof and shall contain provisions for draining.

10.2.3.2 The doors shall seal tightly without a center mullion which would limit the capacity of the chamber.

10.2.3.3 The environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-7721 should be maintained by using electrical heaters immersed in distilled water in the bottom of the chamber. The heaters must be of sufficient capacity to raise the chamber temperature from 65° F to 86° F in 10 minutes and must be automatically controlled.

10.2.3.4 A temperature indicator must be provided. Provisions for continuous recording of temperature and humidity will be provided.

10.2.3.5 The chamber shall operate on 110-volt, 60-cycle alternating current supplied through a single plug-in extension carried with the facility.

10.2.3.6 The velocity of air throughout the test chamber test area shall not be in excess of 25 feet per minute.

10.2.3.7 A minimum air velocity of 900 feet per minute is required to pass over the wet bulb to give accurate reading of the wet bulb temperature.

10.2.3.8 Figure 1 is for reference only and in no way limits or restricts the freedom of the contractor in designing to meet the requirements specified herein. The facility shall generally conform to Figure 1 and provide all of the features indicated therein. This does not exclude the use of a chest type (top opening) chamber.

10.2.4 Construction. The chamber shall be constructed in accordance with the highest grade shop and construction practices.

10.2.5 Performance. This chamber shall be capable of establishing, maintaining, and withstanding all environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-7721, paragraph 4.7.18.1.

10.2.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

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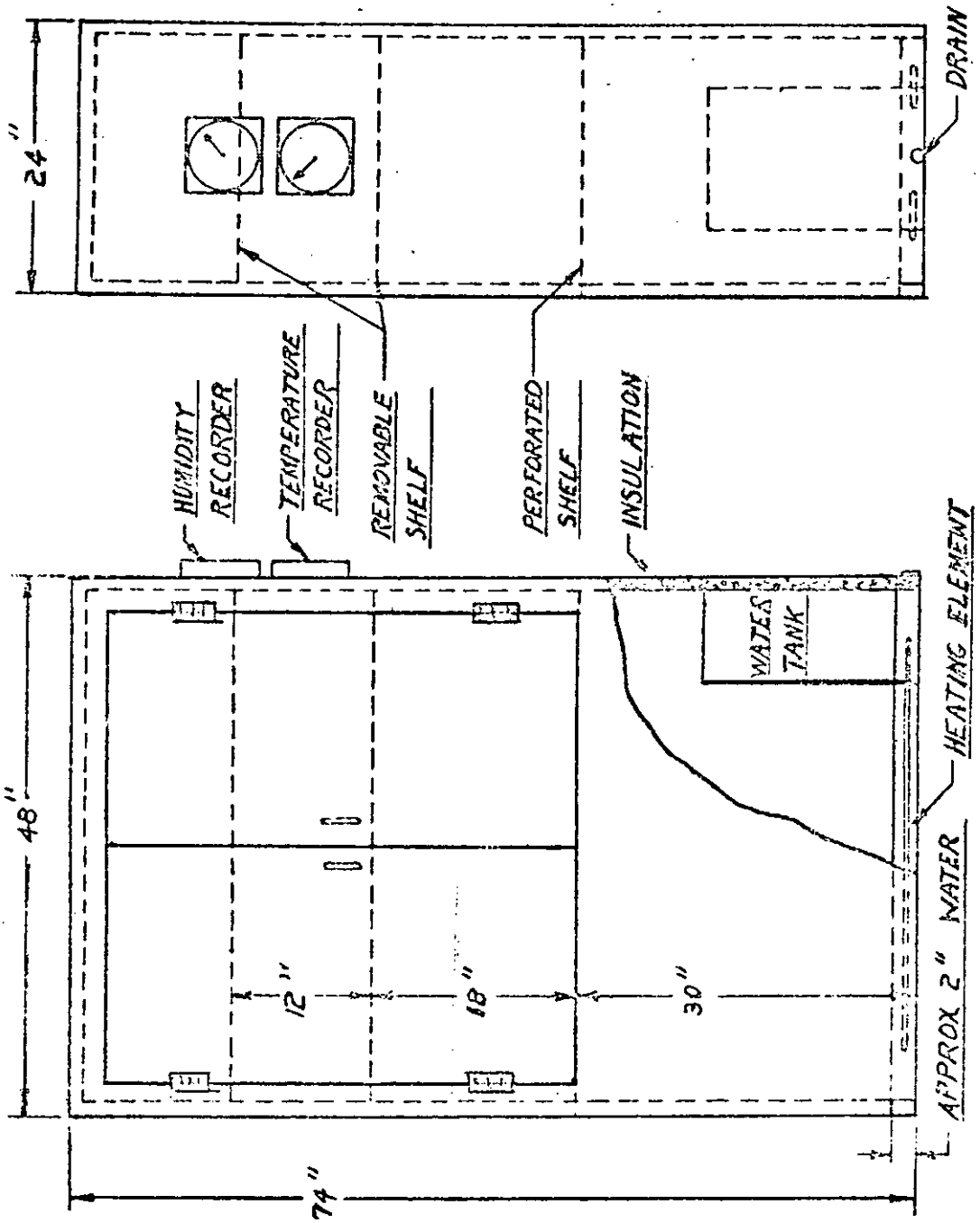


FIGURE 1

DIMENSIONS IN INCHES

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APPENDIX B

20. SALT SPRAY TEST PROCEDURE

20.1 Scope. The purpose of this Appendix is to provide the procedure for salt spray test required by MIL-I-7721, paragraph 4.7.19.1. These procedures are copied from FED-STD-151A, Method 811.1 which is no longer administratively acceptable because it has been superseded. Method 811.1 of FED-STD-151A is desirable for use with MIL-I-7721 although it is not included in the current revision, FED-STD-151B, therefore it is identified and supplied here as a separate entity. (reference paragraph 6.5)

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20.1 SCOPE

20.1.1 This method covers procedures for an accelerated-corrosion test in which specimens are exposed at a temperature of 95° F. (35° C.) to a fine mist of 5 percent or 20 percent sodium chloride solution. When no solution concentration is specified, the 5-percent solution shall be used. The method constitutes an arbitrary performance test which is useful in the establishment and maintenance of certain standards of quality, although it does not closely simulate any common conditions of service. (For Synthetic Sea-Water Spray Test, see method 812.)

20.1.2 The method is applicable to the testing of assemblies, metallic coatings, organic and inorganic coatings on metals, and many nonmetallic materials. The method describes the required conditions for conducting the test for specification purposes. It does not prescribe the type of test specimen, or length of exposure periods to be used for any specific product, or the interpretation to be given to the results. It is desirable in the specification of this test, that the significant surfaces for testing, when applicable, be indicated on the drawing, or in the detail specification, contract, or order. The size and other construction details of the apparatus are optional, provided the conditions obtained meet the requirements of the method described herein.

20.2 APPARATUS

20.2.1 Apparatus used in the salt spray test shall include the following:

(*a*) Exposure chamber with racks for supporting specimens.

(*b*) Salt solution reservoir.

(*c*) Means for atomizing salt solution including suitable nozzles and compressed air supply.

(*d*) Means for heating the exposure chamber and controlling its temperature.

(*e*) Means for cleaning the air and humidifying it at a temperature above that of the exposure chamber.

20.2.1.1 *Exposure chamber.*—The chamber and all accessories shall be made of material such as glass, hard rubber, plastic, or suitably coated wood which will not affect the corrosiveness of the fog. In addition, all parts that come in contact with test specimens shall be made of materials that will not cause galvanic or electrolytic corrosion. The chamber and accessories shall be constructed and arranged as follows:

(*a*) There shall be no direct impinging of the spray, or dripping of the condensate on the specimens.

(*b*) The spray shall circulate freely about all specimens to the same degree.

(*c*) No liquid which has entered the exposure chamber and has come into contact with the test specimens shall return to the salt-solution reservoir.

(*d*) The chamber shall be properly vented.

20.2.1.2 *Atomizers.*—The atomizer or atomizers used shall be of such design and construction as to produce a finely divided, wet, dense fog.

20.2.1.3 *Air supply.*—The compressed air entering the atomizers shall be essentially free from all impurities such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely divided dense fog with the atomizer or atomizers used. To avoid any substantial difference between the salt concentration in the fog particles and that of the salt solution before atomization, the atomizing air must have a relative humidity between 95 and 98 percent for the 5-percent solution and of at least 85 percent for the 20-percent solution after release inside the cabinet. Thus, it is necessary to saturate the air at a higher temperature and cool it to

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95° F. (35° C.) in the chamber in order to comply with the requirement for the NaCl content of the collected solution in 4.2.2.1. A satisfactory method is to pass the air in very fine bubbles through a tower containing water maintained at the proper temperature. Temperatures usually found to be satisfactory for various values of air pressure are given in the following table:

Air pressure (p.s.i.)	Temperature			
	5-percent solution		20-percent solution	
	°F.	°C.	°F.	°C.
12	114	46	110	46
14	117	47	117	47
16	119	48	119	48
18	121	49	121	49

20.2.1.4 *Salt solution.*—The salt used shall be flake salt, substantially free of nickel and copper and containing on the dry basis not more than 0.1 percent of sodium iodide and not more than 0.3 percent of total impurities. "Butter grade" flake salt has been found to meet these requirements. The 5-percent solution shall be prepared by dissolving 5±1 parts by weight of salt in 95 parts by weight of distilled water, or deionized water containing not more than 200 parts soluble solids per million. The 20-percent solution shall be prepared by dissolving 20±2 parts by weight of salt in 80 parts by weight of distilled, or deionized water containing not more than 200 parts soluble solids per million. The solution shall be kept free from solids by filtration or decantation. The solution shall be adjusted to and maintained at a specific gravity of from 1.0268 to 1.0413 for the 5-percent solution and 1.126 to 1.157 for the 20-percent solution when measured at a temperature of 95° F. (35° C.).

20.2.1.4.1 The pH of the salt solution shall be so maintained that solution atomized at 95° F. (35° C.) and collected by the method described in 4.2.2.1 will be in the pH range of 6.5 to 7.2. The pH measurement shall be made immediately after the collected sample is removed from the exposure chamber. The pH shall be measured electrometrically using a glass electrode with a saturated potassium chloride bridge; or colorimetrically using bromthymol blue as indicator. Only diluted c.p. hydrochloric acid of c.p. sodium hydroxide shall be used to adjust the pH.

20.2.1.4.2 When the pH of a salt solution is adjusted at room temperature and atomized at 95° F. (35° C.) the pH of the collected solution will be higher than that of the original solution due to the loss of carbon dioxide at the higher temperature. The following three methods have been found satisfactory for adjusting the pH of the original solution so that the pH of a sample collected after atomization at 95° F. (35° C.) will be within the range of 6.5 to 7.2.

(*a*) Prepare the salt solution at room temperature and adjust the pH so that a 50-ml. sample will have a pH between 6.5 and 7.2 after being boiled gently for 30 seconds and cooled to room temperature. The pH of the sample should be measured immediately after it is cooled. It is necessary for the pH of the original solution to be below 6.5 if this method is followed.

(*b*) Heat the salt solution to boiling and cool to 95° F. (35° C.) or maintain the solution at 95° F. (35° C.) for approximately 48 hours before adjusting the pH to the required range of 6.5 to 7.2.

(*c*) Heat the water from which the salt solution is to be prepared to 95° F. (35° C.) to expel the carbon dioxide before preparing the solution and maintain this temperature while adjusting the pH to within the range of 6.5 to 7.2.

20.3 PREPARATION OF SPECIMENS

20.3.1 Specimens shall be given a minimum of handling, particularly on the significant surfaces, and shall be prepared for test immediately before exposure. Specimens should not be handled with bare hands.

20.3.2 Unless otherwise specified, uncoated metallic specimens and metallic-coated specimens

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shall be thoroughly cleaned of oil, dirt, and grease as necessary until the surface is free from water break. The cleaning methods shall not include the use of corrosive solvents nor solvents which deposit either corrosive or protective films, nor the use of abrasives other than a paste of pure magnesium oxide. Specimens having an organic coating shall not be solvent cleaned.

20.3.3 Unless otherwise specified, cut edges and surfaces not required to be tested and the portions of specimens that come in contact with the support shall be protected with a suitable coating of wax or similar substance impervious to moisture.

20.4 PROCEDURE

20.4.1 **Location of specimens.**—Unless otherwise specified, flat specimens, and other specimens where practicable, shall be supported in such a position that the significant surface is approximately 15° from the vertical, and parallel to the principal direction of horizontal flow of the fog through the chamber. Other specimens shall be positioned as specified in detail specifications or by the purchaser. Rotating specimen holders may be used provided they meet the requirement of this paragraph except in regard to the position of the specimen with respect to the direction of flow of the fog through the chamber. Whenever practicable the specimens shall be supported from the bottom or from the side. When specimens are suspended from the top, suspension shall be by means of nylon cord, glass hooks, or waxed string; the use of metal hooks is not permitted. Specimens shall be positioned so that they do not contact each other or any metallic material or any material capable of acting as a wick, so that they are not shielded from the freely settling fog, and so that corrosion products and condensate from one specimen do not fall upon another specimen or return to the solution reservoir.

20.4.2 **Operating conditions.**

20.4.2.1 *Temperature.*—The test shall be conducted with a temperature in the exposure zone maintained at 95° F. plus 2° F., minus 3° F. (35° C. plus 1° C., minus 1.7° C.). Satisfactory methods for controlling the temperature accurately are by housing the apparatus in a properly controlled constant-temperature room, by thoroughly insulating the apparatus and preheating the air to the proper temperature prior to atomization, or by jacketing the apparatus and controlling the temperature of the water or of the air used in the jacket. No surface in the chamber shall become overheated enough to cause evaporation of the fog followed by condensation of that moisture on the other fog droplets, thereby diluting them. The use of immersion heaters within either (1) the chamber for the purpose of maintaining the temperature within the exposure zone, or (2) in the salt-solution reservoir, is prohibited except that suitable heating units may be used in larger exposure chambers provided they do not cause the temperature within the exposure zone to vary outside the permissible range of operating temperature.

20.4.2.2 *Atomization.*

20.4.2.2.1 The conditions maintained in all parts of the exposure zone shall be such that a suitable receptacle placed at any point in the exposure zone will collect from 0.75 to 2.0 ml. of solution per hour for each 80 sq. cm. of horizontal collecting area (10 cm. dia.) based on an average run of at least 16 hours. When the 5-percent solution is used, the solution thus collected shall have a sodium-chloride content of from 4 to 6 percent (sp. gr. of from 1.0268 to 1.0413 when measured at a temperature between 93° and 97° F. (34° and 36° C.) and a pH from 6.5 to 7.2. When the 20-percent solution is used, the solution thus collected shall have a sodium-chloride content of from 18 to 22 percent (sp. gr. of from 1.126 to 1.157 when measured at a temperature between 93° and 97° F.) (34° and 36° C.) and a pH of from 6.5 to 7.2. At least two clean fog-collecting receptacles shall be used; one placed at near to any nozzle and one as far from all nozzles as any specimen. Receptacles shall be placed so that they are not shielded from the freely settling fog and so

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that no drops of solution from specimens or from other sources will be collected.

20.4.2.2.2 Nozzles shall be made of material nonreactive to the salt solution. Glass or methyl-methacrylate (Lucite) nozzles with nylon filters are suggested. Suitable atomization has been obtained in boxes having a volume of less than 12 cubic feet with the following conditions:

(*a*) Nozzle pressure of 12 to 16 pounds per square inch.

(*b*) Orifices of from 0.02 to 0.03 inch in diameter.

(*c*) Atomization of approximately three quarts of the salt solution per 10 cubic feet of box volume per 24 hours.

20.4.2.2.3 When using boxes having a volume considerably in excess of 12 cubic feet, the above conditions may have to be modified in order to meet the requirements for operating conditions.

20.4.3 **Length of test.**—The length of the salt spray test shall be as required by the detail specification or drawing. Unless otherwise specified, the test shall be run continuously for the time indicated or until definite indication of failure is observed, with interruption only for adjustment of the apparatus and inspection of the specimen. Operations shall be scheduled so that interruptions are held to a minimum. The following exposure periods are convenient for scheduling tests and are recommended for use in detail specification: 2, 4, 6, 8, 16, 24, 32, 48, 72, 96, 200, 240, 500, or 720 hours. When a retest is authorized for a sample or samples that failed to pass a specified test, the retest shall be run for the required number of hours with a minimum of interruptions or opening of the cabinet.

20.4.4 **Examination of specimens.**—At the end of the test, the specimens shall be examined immediately to determine compliance with the standard of acceptability specified in the detail specification or drawing. If necessary to aid in examination, a gentle wash or dip in clean running water not warmer than 100° F. (38° C.) and blowing with clean dry air may be used. Examination at stated intervals during the test may be required.

20.5 REPORT OF RESULTS

20.5.1 The following information shall be recorded, unless otherwise prescribed in the detail specification or drawing for the material or product being tested.

(*a*) Solution concentration.

(*b*) Type of salt and water used in preparing the salt solution.

(*c*) All readings of temperature within the exposure zone of the chamber.

(*d*) Record of daily test made on each fog collecting device including the following:

(1) Volume of salt solution collected in milliliters per hour per 80 sq. cm. of horizontal collecting area.

(2) Sodium chloride content or specific gravity (including temperature at which sp. gr. measurement was made) of the collected solution.

(3) pH of collected solution.

(*e*) Type of specimen and its dimensions, or number or description of product or part tested.

(*f*) Method of supporting specimen in the salt spray chamber.

(*g*) Description of protection used on cut edges and other surfaces not required to be tested.

(*h*) Length of exposure period.

(*i*) Cause and length of interruptions during test.

(*j*) Results of all inspections.

(*k*) Method of specimen surface preparation.

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